# A generic energy-balance solver for precooled Joule–Thomson cryocoolers: theory, implementation, and validation against Zhang & Dang (2021) and other hybrid coolers

Companion document to T. Zhang and H. Dang, *Cryogenics* **116** (2021) 103282

**Abstract**

This document is written to be read next to Zhang & Dang's paper on a He-3 Joule–Thomson (JT) cooler precooled by a four-stage pulse-tube cooler. It does four things. (i) It restates the paper's physical model, with every assumption made explicit, and derives the results the paper states (the last-stage optimum $\Delta h_T = q_L$, the effect of exchanger efficiency, the cascade balance) from the first law. (ii) It describes the solver built from that theory: the mapping from each equation to a function, the property tables and interpolation, the choked-flow model and the optimisation logic. (iii) It validates the solver against all quantitative results of the paper (27 of 27 checks, e.g. $P_{U,\mathrm{opt}} = 0.358$ MPa vs. 0.36 MPa, $Q_C = 18.62$ mW vs. 18.7 mW) and against five cases from two other papers with the same architecture, with honest account of where and why the ideal model departs from measurements. (iv) It shows how the validated solver is used to *design* a coupled hybrid JT cooler: for a chosen cold end it returns the optimum operating pressure and last-stage precooling temperature, and maps the temperature and heat load required of every upstream precooler, which is the information needed to select a precooling cryocooler. Everything is driven from one command-line script, so every number in this document can be regenerated.

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# 1 Introduction and plan

A Joule–Thomson (JT) cooler produces refrigeration by throttling a high-pressure gas through a restriction; the temperature fall, and above all the refrigeration it can absorb at the cold end, depend strongly on the temperature the gas has before the restriction. Below about 4 K in He-4 and about 3 K in He-3 the gas has to be precooled to a few kelvin before throttling gives any useful effect. A *hybrid* (coupled) JT cooler does this with a cascade: the gas passes through a chain of counterflow heat exchangers (CHEX) in which the returning low-pressure vapour cools it, and between successive CHEXs a precooling heat exchanger (PHEX) is thermally anchored to a stage of a mechanical cryocooler (here a multi-stage pulse-tube cooler, SPTC). Zhang & Dang analyse such a cooler for He-3 at 1 K and give design rules.

**Aim of this work.** Build a single solver that (a) reproduces the paper from its own assumptions, (b) is general enough to solve other published coolers with the same architecture, and (c) can be used to design a new one. The route followed, and the structure of this document, is:

1. Understand the paper's theory and write down its assumptions (Sections 2–3).
2. Decide how to turn it into a solver, and which choices were made and why (Section 4).
3. Trace how the solver evolved from the handwritten derivation, with the decisions and cases that shaped it (Section 5).
4. Map theory to code (Sections 6–7).
5. Validate on the paper itself (Section 8).
6. Validate on other published coolers (Section 9).
7. Use it for design (Section 10), then limitations and reproducibility.

# 2 System and notation

## 2.1 Architecture

Figure 1 shows the architecture used by the paper (Fig. 1 there) and by every case in this document, for $N$ CHEXs and $N-1$ PHEXs; the paper has $N=4$. High-pressure gas at $P_U$ enters the warm end of CHEX 1 at $T_H$ (300 K). It leaves CHEX $i$, is cooled by PHEX $i$ to $T_{\mathrm{pre},i}$, and enters CHEX $i+1$. After the last CHEX it is throttled through the orifice (isenthalpic) to the low pressure $P_D$, partly evaporates at $T_C$ in the evaporator (where the refrigeration $Q_C$ is absorbed), and the saturated vapour returns through all CHEXs, cooling the incoming stream, to leave at the warm end.

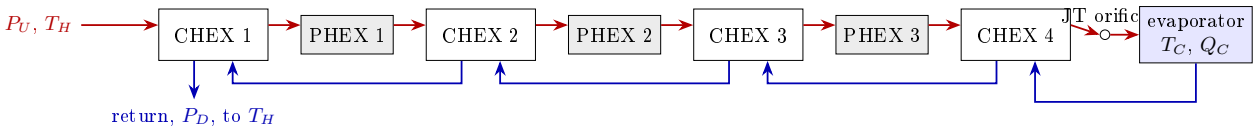



**Figure 1:** Architecture: red = high-pressure supply stream, blue = low-pressure return stream (schematic; cf. Fig. 1 of Zhang & Dang). PHEX $j$ removes heat $Q_{\mathrm{pre},j}$ from the supply stream and is anchored to a stage of the precooling cryocooler.

## 2.2 State points

We follow the paper's numbering (its Fig. 2) for $N=4$. High-pressure side: 1 (CHEX 1 hot inlet, $T_H$), 2 (CHEX 1 hot outlet = PHEX 1 inlet), 3 (PHEX 1 outlet = CHEX 2 inlet, $T_{\mathrm{pre},1}$), 4, 5 ($T_{\mathrm{pre},2}$), 6, 7 ($T_{\mathrm{pre},3}$), 8 (CHEX 4 hot outlet, before the orifice). 9 is after the orifice (two-phase, $h_9 = h_8$), 10 is saturated vapour leaving the evaporator, 11, 12, 13 are the return-stream states between CHEXs and 14 is the warm-end exit. Points 1D, 3D, 5D, 7D are the states on the low-pressure isobar at the temperatures of points 1, 3, 5, 7. The code uses exactly this labelling for $N=4$ and generalises it for other $N$ (Appendix A); Fig. 2 shows the resulting cycle in the $P$–$h$ plane.

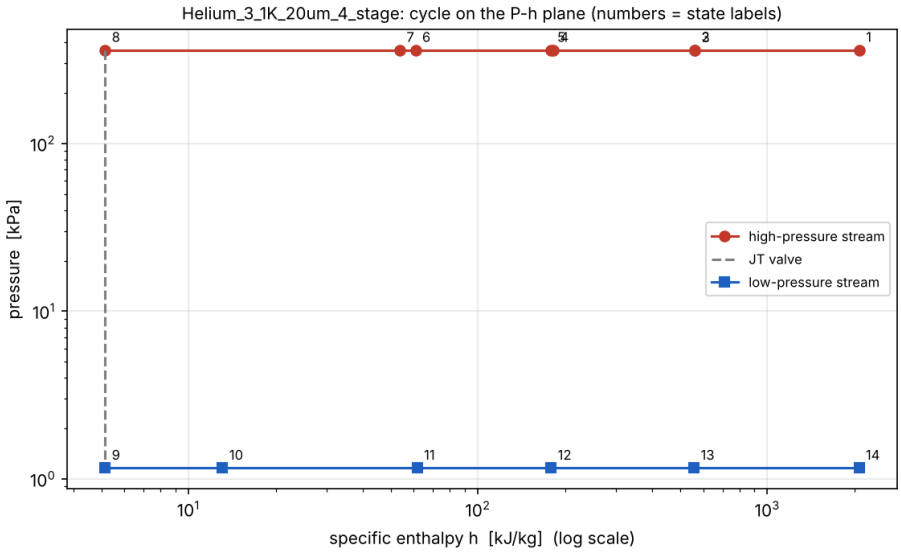



**Figure 2:** Cycle of the design point of the paper ($T_C = 1$ K, $P_U = 0.358$ MPa, $T_{\mathrm{pre}} = (25.6, 12.3, 8)$ K, $d = 20$ µm) on the $P$–$h$ plane (log–log), with the state labels. Output of `jt_run.py run`; see Section 10.

### 2.3 Symbols

| | |
|---|---|
| $h,\ s,\ \rho$ | specific enthalpy, entropy, density |
| $P_U,\ P_D$ | high pressure; low pressure $= P_{\mathrm{sat}}(T_C)$ |
| $T_C,\ T_H$ | evaporator (cold-end) temperature; warm-end temperature (300 K) |
| $T_{\mathrm{pre},j}$ | gas temperature leaving PHEX $j$ (at $P_U$) |
| $\eta_i$ | effectiveness of CHEX $i$ (Section 3.3) |
| $q,\ Q$ | per unit mass flow [kJ/kg], and power $Q = \dot{m}q$ [mW] |
| $q_C,\ Q_C$ | refrigeration at the evaporator |
| $q_{\mathrm{pre},j},\ Q_{\mathrm{pre},j}$ | heat removed at PHEX $j$ |
| $\Delta h_T$ | isothermal JT enthalpy difference $h(P_D, T_{\mathrm{pre,last}}) - h(P_U, T_{\mathrm{pre,last}})$ |
| $q_L$ | latent limit $h_g(T_C) - h(P_U, T_C)$ |
| $G,\ d$ | orifice mass flux [kg m$^{-2}$s$^{-1}$]; orifice diameter |

## 3 Theory: the physical model and its assumptions

### 3.1 Assumptions, stated once

All results below follow from the following assumptions. Items marked $*$ are the paper's; the others are choices we made explicit because they matter when comparing with other work.

A1$^*$ Steady state; one-dimensional counterflow; the two streams carry the same mass flow $\dot{m}$ (no leaks between them, no mass storage).

A2$^*$ No heat in-leak from the surroundings and no axial conduction along the exchangers. All heat entering the system enters at the evaporator ($Q_C$) or leaves at the PHEXs.

A3$^*$ No pressure drop along either stream: the high-pressure stream is at $P_U$ everywhere, the low-pressure stream at $P_D$ everywhere.

A4$^*$ The orifice is adiabatic: $h_9 = h_8$.

A5$^*$ The evaporator is at saturation: the stream leaving it is saturated vapour at $T_C$, and $P_D = P_{\mathrm{sat}}(T_C)$ (zero quality-independent temperature gradient in the evaporator).

A6$^*$ The precooler delivers the gas to a specified temperature $T_{\mathrm{pre},j}$; how the cryocooler achieves this (its own capacity curve) is outside the model.

A7 Each CHEX is characterised by one number, the effectiveness $\eta_i$, defined from an enthalpy balance, not by $UA$ and length. Properties vary through the exchanger as the real fluid dictates (no ideal-gas shortcuts).

A8 The mass flow is set by the orifice (choked flow of the real fluid), or given from a measurement. It is never obtained from a compressor model.

A9 No discharge-coefficient correction ($C_d = 1$) and no multi-orifice effects.

Assumptions A2, A3 and A9 are optimistic and are why a real cooler delivers less than the model (Sections 9 and 11).

### 3.2 Energy balance of one counterflow exchanger

Label the four ends of a CHEX: hot stream in $A$ (at $P_U$), hot stream out $B$; cold stream in $C$ (at $P_D$), cold stream out $D$. With A1–A3 the first law for each stream gives the same heat duty $Q$ per unit mass flow:

$$h_B = h_A - Q, \qquad h_D = h_C + Q. \tag{1}$$

Two physical limits bound $Q$. The *warm-end limit*: the cold stream cannot leave warmer than the hot stream enters, so $h_D \le h(P_D, T_A)$, i.e.

$$Q \le Q_{\text{warm}} = h(P_D, T_A) - h_C. \tag{2}$$

The *cold-end limit*: the hot stream cannot leave colder than the cold stream enters, $h_B \ge h(P_U, T_C)$ with $T_C$ now the temperature of state $C$:

$$Q \le Q_{\text{cold}} = h_A - h(P_U, T_C). \tag{3}$$

$Q_{\max} = \min(Q_{\text{warm}}, Q_{\text{cold}})$. This is the enthalpy (not temperature) form of the effectiveness definition and it is essential for real gases: because the specific heat of the high-pressure stream differs strongly from that of the low-pressure stream (Fig. 3), the two limits are generally different and *which one binds is a result, not an assumption.* For He-3 at 0.36 MPa, $c_p$ rises to about 12 kJ/kg/K near 5 K and falls to about 1.3 kJ/kg/K at 1 K, while at 1.16 kPa $c_p \approx 7$ kJ/kg/K throughout; the high-pressure stream has the larger capacity rate above $\approx 4$ K and the smaller one below it, which is why the last CHEX is cold-end limited and the others are warm-end limited (this is the "pinch" discussion of the paper, its Fig. 4).

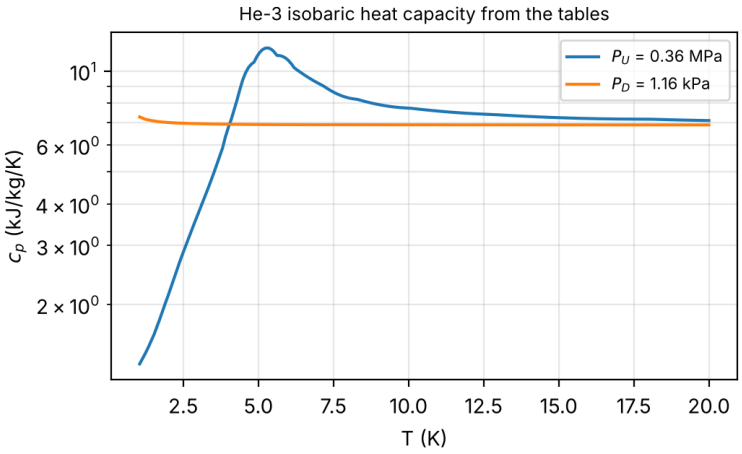



**Figure 3:** He-3 isobaric heat capacity from the tables at $P_U = 0.36$ MPa and $P_D = 1.16$ kPa (cf. Fig. 4 of the paper). Computed from `h(P,T)` by finite differences.

### 3.3 Effectiveness: the paper's definition and ours

The paper defines the effectiveness of the last CHEX from the cold side only (its Eq. 4):

$$\eta_4 = \frac{h_{11} - h_{10}}{h_{7D} - h_{10}}. \tag{4}$$

This is $Q/Q_{\text{warm}}$. We define instead

$$\eta = \frac{Q}{Q_{\max}}, \qquad Q_{\max} = \min(Q_{\text{warm}}, Q_{\text{cold}}). \tag{5}$$

The two coincide whenever the exchanger is warm-end limited, and for the paper's last CHEX at and below the optimal $P_U$ (the regime of its analysis). When $P_U$ is above the optimum the last CHEX is cold-end limited, $Q_{\text{cold}} < Q_{\text{warm}}$, and Eq. (4) with $\eta < 1$ would permit $Q > Q_{\text{cold}}$, i.e. $T_B < T_C$: the hot stream leaves colder than the cold stream enters, which is unphysical. Definition (5) is therefore used throughout,

and we state this as a deliberate, checkable difference from the paper (it does not change any number in the paper's regime; see the validation in Section 8). The same definition makes $\eta_i$ directly comparable with the effectiveness quoted by other authors from measured temperatures (Section 3.9).

### 3.4 The last stage: $q_C = \min(\Delta h_T,\, q_L)$

Consider the last CHEX, orifice and evaporator (states 7–11, paper Fig. 3). The refrigeration absorbed per unit mass is the enthalpy rise in the evaporator,

$$q_C = h_{10} - h_9 = h_{10} - h_8, \tag{6}$$

with $h_{10} = h_g(T_C)$ and $h_8 = h_7 - Q$ from Eq. (1). For an ideal CHEX ($\eta = 1$) $Q = Q_{\max}$ and Eq. (6) becomes

$$\text{warm-end limited: } q_C = h_{10} - h_7 + (h_{7D} - h_{10}) = h(P_D, T_7) - h(P_U, T_7) \equiv \Delta h_T, \tag{7}$$

$$\text{cold-end limited: } q_C = h_{10} - h(P_U, T_C) \equiv q_L. \tag{8}$$

Because the limiting case is the one with the smaller $Q$, which is the one giving the larger $h_8$ and hence the *smaller* $q_C$,

$$q_C = \min(\Delta h_T,\, q_L), \tag{9}$$

which is the paper's Eqs. (30)–(31). $\Delta h_T$ is the isothermal JT enthalpy difference at the precooling temperature (how much enthalpy the gas gives up when throttled isothermally at $T_7$); $q_L$ is the latent heat of the liquid at the evaporator temperature compressed to $P_U$ (the most the liquid could ever absorb on evaporation).

**Dependence on pressure.** From $(\partial h/\partial P)_T = v(1 - \beta T)$ ($\beta$ the volume expansivity),

$$\Delta h_T(P_U) = \int_{P_D}^{P_U} v\,(\beta T - 1)\,\mathrm{d}P, \qquad q_L(P_U) = h_g - h(P_U, T_C), \quad \frac{\mathrm{d}q_L}{\mathrm{d}P_U} = -v_L(1 - \beta_L T_C) < 0. \tag{10}$$

For the liquid $\beta_L T_C \ll 1$, so $q_L$ falls almost linearly with $P_U$ and vanishes at a maximum pressure $P_{U,\max}$ (1.131 MPa for He-3 at 1 K; paper: 1.13 MPa), above which no refrigeration is possible at any precooling temperature. $\Delta h_T$ rises with $P_U$ (below the inversion temperature) and saturates. Hence $q_C(P_U)$ increases with $\Delta h_T$ until the two curves cross, then decreases along $q_L$: the maximum is at the crossing,

$$\Delta h_T(P_{U,\mathrm{opt}}, T_{\mathrm{pre}}) = q_L(P_{U,\mathrm{opt}}), \tag{11}$$

the paper's Eq. (34), and equality of the two is both necessary and sufficient for the optimum (Fig. 4, left). Equation (11) is one equation in two unknowns, so it defines a curve $P_{U,\mathrm{opt}}(T_{\mathrm{pre}})$ (paper's Fig. 8, fitted by its Eqs. 35–36).

**Dependence on $T_{\mathrm{pre}}$.** $\Delta h_T$ has a maximum in temperature (near 4 K for He-3 at 0.36 MPa, dashed green curve in Fig. 4, right) and falls strongly as $T_{\mathrm{pre}}$ rises; $q_L$ does not depend on $T_{\mathrm{pre}}$. While $\Delta h_T > q_L$ the refrigeration is on the plateau $q_C = q_L$: cooling the gas further does not help. The *warmest* $T_{\mathrm{pre}}$ on the plateau is therefore the best choice, $T_{\mathrm{pre,opt}}$: it delivers the same $q_C$ with the least demanding precooler. At $P_U = 0.358$ MPa this is 8 K, which is the paper's choice of $T_{\mathrm{pre3}} = 8$ K.

**Imperfect last CHEX.** For $\eta < 1$, $Q = \eta Q_{\max}$ and Eq. (6) gives

$$q_C = \eta\, Q_{\max} - (h_7 - h_{10}) = \eta\, \min(\Delta h_T, q_L) - (1 - \eta)\,(h_7 - h_{10}). \tag{12}$$

The penalty is $(1 - \eta)$ times the *whole* enthalpy difference between the warm hot-stream end and the cold return stream (here $h_7 - h_{10} = 40.7$ kJ/kg), about five times the useful $q_C$. Hence a 1 % loss of effectiveness costs about 6 % of $q_C$ (we find 7.96 → 7.47 kJ/kg for $\eta = 1 \to 0.99$), which is why the last CHEX is the critical component. Together with the associated mass-flow change (Section 3.6), this gives the large sensitivity of $Q_C$ shown in Section 8.

### 3.5 Cascade balance and what the upstream precoolers can change

For ideal exchangers the paper obtains, from Eq. (1) applied to each CHEX and the precoolers (its Eqs. 18–22),

$$q_C = \sum_j q_{\mathrm{pre},j} + \big[h(P_D, T_H) - h(P_U, T_H)\big], \tag{13}$$

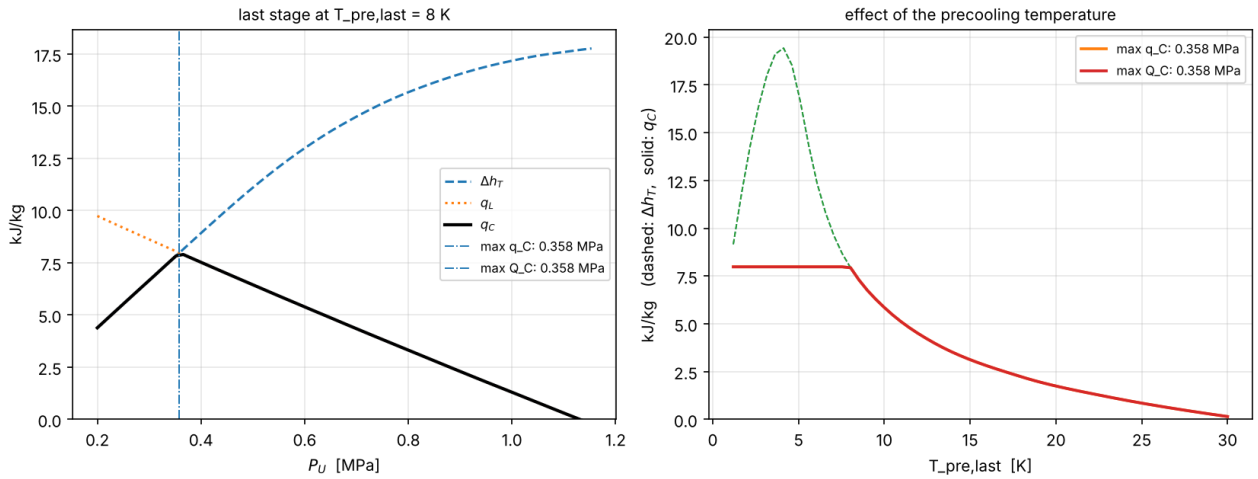



**Figure 4:** Last stage of the design example. Left: $\Delta h_T$, $q_L$ and $q_C = \min$ versus $P_U$ at $T_{\mathrm{pre,last}} = 8$ K; the optimum is the crossing at 0.358 MPa and the curve reaches zero at 1.13 MPa (cf. paper Fig. 5). Right: effect of $T_{\mathrm{pre,last}}$ (at the optimal $P_U$ the plateau $q_C = 7.98$ kJ/kg ends at 8 K; cf. paper Fig. 7).

where the bracket is the (negative) isothermal JT enthalpy difference at ambient, and $Q_C = \dot{m} q_C$. Equation (13) is just the first law over the whole machine: everything the precoolers remove, plus the net enthalpy the compressed gas brings in, balances the refrigeration. For $\eta < 1$ the same balance holds with $h(P_D, T_{D1})$ in place of $h(P_D, T_H)$, $T_{D1}$ being the actual warm-end return temperature; the solver evaluates it that way and checks it.

**An important consequence.** The last CHEX sees fixed boundary states on both streams: its hot inlet $(T_{\mathrm{pre,last}}, P_U)$ and its cold inlet (saturated vapour at $T_C$). Its duty, $q_C$, $T_8$, and through $T_8$ the orifice flux and $\dot{m}$, are therefore determined by $T_C$, $P_U$, $T_{\mathrm{pre,last}}$, $\eta_{\mathrm{last}}$ and the orifice alone. *The upstream precooler temperatures cannot change $q_C$, $\dot{m}$ or $Q_C$*; they only decide how the (fixed) total precooling load $\sum_j q_{\mathrm{pre},j}$ in Eq. (13) is distributed among the PHEXs. This separates the design problem into two stages: optimise the last stage (Section 3.7), then choose the upstream temperatures by a different criterion (Section 3.8). We verified the invariance numerically (Fig. 10: the total load and $Q_C$ stay constant while any one upstream temperature is swept over its full range).

## 3.6 Mass flow: choked flow through the orifice

$Q_C = \dot{m} q_C$ needs $\dot{m} = \pi d^2 G/4$, with $G$ the mass flux through the orifice. The paper computes $G$ for real He-3 from the choked-flow condition using the fundamental derivative-based factor $\Gamma$ and the real-gas equation of state (its Eqs. 41–48), and fits the results (Table 2) with an algebraic form (Eq. 49) as function of $P_U$ and $T_8$. We use an equivalent but simpler formulation that needs only the property tables: a *homogeneous-equilibrium model* (HEM). For stagnation state $(P_0, h_0, s_0)$ (state 8) and an isentropic expansion to pressure $P$,

$$G(P) = \rho(P, s_0)\sqrt{2\left[h_0 - h(P, s_0)\right]}, \qquad G_{\mathrm{choked}} = \max_{P_D \le P < P_U} G(P). \tag{14}$$

The maximum is found on 500 logarithmically spaced pressures and refined by golden-section search. If the maximum lies at $P = P_D$ the flow is not choked (flag `UNCHOKED_RESTRICTION_FLOW`); this is typical when the inlet is liquid-like. Equation (14) agrees with the paper's real-gas results to 0.25–0.72 % (mean) at $T_C = 1, 2, 2.6, 3$ K (Table 1). The paper's own fit (Eq. 49) is also implemented (`--flow-model paper_fit`) and is itself reproduced by our HEM data to 0.2–1.3 % in the worst case.

**Why $\dot{m}$ depends on $\eta$.** $G$ depends on the stagnation state 8; with $\eta < 1$, $T_8$ rises (from 1 K to 1.91 K at $\eta = 0.97$), the fluid is less dense and its flux falls, hence the decrease of $\dot{m}$ with $\eta$ in the paper's Fig. 15 (2.339 → 2.267 mg/s here).

## 3.7 Optimising $Q_C$ rather than $q_C$

Since $Q_C = \dot{m} q_C$ and $\dot{m}$ increases with $P_U$ (flux grows with upstream pressure), the pressure for maximum $Q_C$ is above that for maximum $q_C$ *when the flux gain outweighs the $q_C$ loss*. In the paper's case with $T_{\mathrm{pre}} = 5$ K

the $q_C$ maximum is at 0.21 MPa but the $Q_C$ maximum is at 0.36 MPa (paper, Sect. 5); we obtain 0.213 and 0.352 MPa (Section 8). A second subtlety is that $q_C$ alone has *no* interior optimum in $T_{\mathrm{pre,last}}$ (it is monotone: colder is never worse), so "the optimum" only exists with a constraint or a different objective. We therefore (i) optimise $Q_C$ when a flow is available, with a "practical" option that picks the warmest $T_{\mathrm{pre}}$ and the lowest $P_U$ whose $Q_C$ is within 0.5 % of the maximum (a flat optimum: Fig. 5), and (ii) use $T_{\mathrm{pre,opt}}$ of Eq. (11) when $T_{\mathrm{pre}}$ is to be chosen.

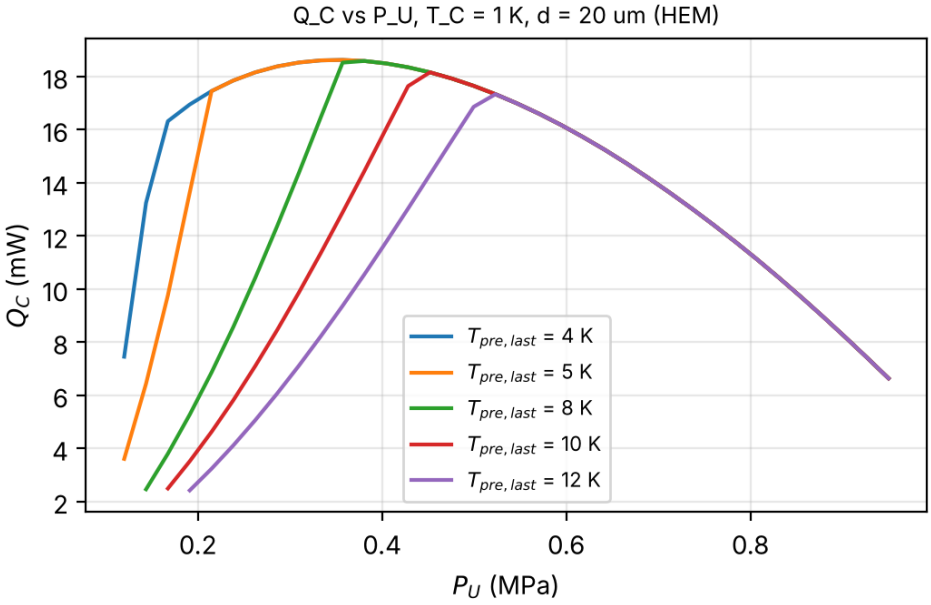



**Figure 5:** $Q_C$ versus $P_U$ at $T_C = 1$ K, $d = 20$ µm (HEM flux) for several $T_{\mathrm{pre,last}}$ (cf. Fig. 13 of the paper). Curves with colder $T_{\mathrm{pre}}$ reach the common envelope (maximum 18.6 mW at about 0.36 MPa) at lower $P_U$; warmer $T_{\mathrm{pre}}$ forces the optimum to higher $P_U$ and lower $Q_C$.

### 3.8 Choosing the upstream temperatures

By Section 3.5 upstream temperatures $T_{\mathrm{pre},j}$ ($j < N-1$) leave $q_C$ unchanged, so another criterion is needed. Physically, a precooler at temperature $T_j$ rejecting $Q_j$ to ambient $T_H$ needs at least the Carnot work $Q_j(T_H/T_j - 1)$, and real cryocoolers are much more demanding at low $T_j$. We therefore minimise

$$\Phi = \sum_j Q_{\mathrm{pre},j}\left(\frac{T_H}{T_j} - 1\right) \tag{15}$$

over the upstream $T_{\mathrm{pre},j}$, subject to $T_H > T_1 > T_2 > \cdots > T_{\mathrm{last}}$ and every $Q_{\mathrm{pre},j} \geq 0$ (a PHEX may not heat the gas). A differential-evolution search is used, because $\Phi$ has non-smooth regions where a PHEX load reaches zero. This is only one defensible criterion (the real choice depends on the capacity curves of the cryocooler used, Section 10), so the code also produces the full map of loads versus each upstream temperature, from which the user can choose any other.

### 3.9 Measured flow and effectiveness from temperatures

Experimental papers give $\dot{m}$ and temperatures rather than an orifice diameter. The solver accepts a measured `mdot_mg_s` instead of $d$ ($d$ and $\dot{m}$ together are rejected as over-specified). If the paper reports the temperatures at the two ends of the last CHEX (e.g. hot stream in at $T_A$, return stream out at $T_D$), the effectiveness follows from Eq. (5) in the warm-limited case,

$$\eta = \frac{h(P_D, T_D) - h_C}{h(P_D, T_A) - h_C}. \tag{16}$$

This is used for the Liu et al. case (Section 9).

### 3.10 Internal pinch check

Eq. (1)–(5) constrain only the ends of an exchanger. With strongly varying $c_p$ the stream temperatures could still cross *inside*. The solver therefore integrates the exchanger at 20 equal enthalpy steps, computing both stream temperatures, and flags a negative gap (`INTERNAL_TEMPERATURE_CROSSOVER`); this is a warning, not an error. In the paper's design CHEX 1 shows a small crossover (minimum gap $-0.01$ K): a sign that the ideal $\eta = 1$ is physically marginal in that exchanger, not a solver failure.

## 4 Plan: from the paper to a solver

The decisions below were taken after discussion and are recorded here because they determine what the solver does and does not claim.

1. **Energy balance over $UA$ models.** The paper works in the $P$–$h$ plane with effectiveness; a $UA$–NTU model would need geometry and heat-transfer correlations that the target papers do not provide. All inputs are therefore thermodynamic: $T_C, P_U, T_{\mathrm{pre},j}, \eta_i, d$ or $\dot{m}$.
2. **Properties from the paper's own tables.** For He-3 the supplied single-phase and saturation tables (not an equation of state) are the single source of truth; this removes the ambiguity of reference states and keeps the reproduction faithful. For other fluids CoolProp is used through the same interface.
3. **Everything generic in $N$.** The cycle is solved for any number of CHEXs, not only four, so other coolers (3 CHEX with 2 PHEXs, etc.) run unchanged.
4. **HEM for the orifice** (reasoned above), with the paper's own fit available for comparison.
5. **Fail loudly.** Inputs outside the tables raise an error (no extrapolation); the feasibility check explains infeasibility rather than returning a number.
6. **Validation first, claims second.** The solver was checked against all quantitative statements of the paper before it was applied to other papers; other-paper cases carry their reported values and a reference block, and mismatches are explained rather than tuned away (no free fitting parameter is used to force agreement).
7. **One front door.** A single command-line script (no GUI) drives validation, other papers and own designs; every run is stored with its configuration so that a result can always be traced to inputs.

## 5 How the solver evolved: from the handwritten derivation to the present code

This section records how the work actually developed, so that each design choice can be traced to the discussion that produced it. It is reconstructed from the handwritten derivation sheets, two earlier working chats (a first assistant session and a planning session), the review session that preceded this one, and this session. Where a reason for a decision is not recorded, none is given.

### 5.1 Stage 0: the handwritten derivation (the starting point)

The handwritten sheets (8 pages) are the origin of the model. They contain:

1. **Objective.** From a required $T_C$ and $Q_C$, find $P_U$, the precooler temperatures $T_i$ and loads $Q_{\mathrm{pre},i}$ so that the pulse-tube cooler and the JT cooler could be designed together. Inputs are $T_C$, $P_U$, $T_{\mathrm{pre},1..3}$, $\eta_{1..4}$.
2. **Hard assumptions.** No pressure loss (states 1–8 at $P_U$, 9–14 at $P_D$); the gas leaves each PHEX at the PHEX temperature ($T_3 = T_{\mathrm{pre},1}$, $T_5 = T_{\mathrm{pre},2}$, $T_7 = T_{\mathrm{pre},3}$); the fluid is He-3.
3. **CHEX.** Four states $A, B, C, D$, a fictitious state $A'$ at $(T_A, P_D)$, effectiveness $\eta = (h_D - h_C)/(h_{A'} - h_C)$ (cold-side definition), hence $h_D = h_C + \eta(h_{A'} - h_C)$ and $h_B = h_A - \eta(h_{A'} - h_C)$, all explicit.
4. **Pinch argument** for CHEX 4 at $\eta \to 1$: $c_{p,U} < c_{p,D}$ gives a cold-end pinch and $c_{p,U} > c_{p,D}$ a hot-end pinch.
5. **Cold end.** $h_8 = h_9$; $q_C = h_{10} - h_8$; with $\eta_4$: $q_C = h_{10} - h_7 + \eta_4(h_{7D} - h_{10})$, $\to \Delta h_T$ at $\eta_4 = 1$. Quality argument: $q_C = (x_{10} - x_9)h_{fg}$, the maximum liquid evaporated is $q_L = (1 - x_9)h_{fg}$, so $q_L \geq \Delta h_T$ is required and in general $q_C = \min[\Delta h_T, q_L]$.
6. **Optimum.** $\Delta h_T$ rises and $q_L$ falls with $P_U$, so $q_C$ has a maximum; triplets $(T_{\mathrm{pre}3}, P_U, q_C)$ form a surface, $P_{U,\mathrm{opt}}$ and $q_{C,\mathrm{opt}}$ are fitted as functions of $T_{\mathrm{pre}3}$; the maximum is at 8 K, 0.36 MPa, $P_D = 1.16$ kPa.

7. **Mass flow.** The valve as a converging–diverging nozzle: ideal-gas flux $G_{ID}$, real flux $G = \Gamma G_{ID}$, throat found from $s^* = s_0$ and $h^* + a^{*2}/2 = h_0$, $\Gamma$ fitted as function of $(P_U, T_{\mathrm{pre3}})$; $Q_C = \frac{\pi d^2}{4}\Gamma G_{ID} q_C$; overall optimum of $Q_C$ over $(T_{\mathrm{pre3}}, P_U)$.

## 5.2 Stage 1: first working chat — architecture

The first chat turned this into a code plan. The decisions made there, and kept to the end, are:

- **Whole cycle, not only states 7–11.** The model covers all states 1–14; precooling loads follow from enthalpy differences across each PHEX ($Q_{\mathrm{pre},3} = \dot{m}(h_6 - h_7)$, etc.). Given all states, the JT side can be read as the load the cryocooler must supply.
- **Reusable CHEX block, generic in $n$.** A CHEX is a component with states $A, B, C, D$; $n$ CHEXs and $n-1$ PHEXs are chained, state numbering generalised from the paper's.
- **Explicit propagation from the cold end.** State 7 follows from $(P_U, T_{\mathrm{pre3}})$, state 10 from $T_C$ (saturated vapour at $P_{\mathrm{sat}}(T_C)$); CHEX 4 gives states 8 and 11, CHEX 3 then has known $A$ (from $T_{\mathrm{pre2}}$) and $C$ (state 11), and so on to CHEX 1.
- **The core is a black box.** The cycle solver is kept separate from a reproduction script that does the sweeps, plots, fits and error metrics; the paper's *published fits* (Eqs. 35, 36, 49) are kept separate from *our own* calculations so that "reconstructing a published curve" is never confused with "reproducing it independently".
- **Tests on a synthetic fluid** check software logic only, never He-3 physics.
- **HEM flow.** The speed of sound is not available for He-3 in CoolProp, so the critical flux is found by maximising $G(P)$ along the isentrope (HEM), kept distinct from the paper's real-gas $\Gamma$ method.
- **The property problem.** Since no He-3 backend exists, the solver was designed so that a CoolProp-like tabular backend could be plugged in; a standard property-request spreadsheet (10 000 $P$–$T$ points, saturation list, property catalogue, backend contract) was prepared for the data provider. Properties could be requested over 0.3–350 K, 0.01 Pa–2 MPa; the tables finally supplied cover 0.5–300 K, 1 Pa–3 MPa.

## 5.3 Stage 2: planning chat — pseudo-code and corrections

A second chat wrote layered pseudo-code (properties → components → flow → cycle → applications → entry point) and a validation plan. Three corrections made there shaped the final solver:

- **No fixed-point iteration is needed.** The enthalpy chain does not depend on $\dot{m}$; the coupling is one-way, $T_C, P_D, T_{\mathrm{pre}}, \eta \to$ states $\to (T_8, P_U) \to G^* \to \dot{m} \to Q_C, Q_{\mathrm{pre,i}}$. Iteration would only appear with pressure drops, parasitic loads or load-dependent precooler temperatures (none included).
- **Both CHEX closures are explicit,** so no root-finding is needed (this is why a solve takes milliseconds).
- **The paper's $\Gamma$ is inconsistently stated** (Eq. 43 gives $G = G_{ID}\Gamma$, the procedure divides $G_{ID}$ by $G$); the physically meaningful form is $\Gamma = G/G_{ID}$ and a solver that computes $G$ directly from real-gas properties sidesteps the ambiguity. This is one more reason for HEM.

It also recorded the validation targets that became the scorecard: round-trip and energy balance, ideal limit $q_C = h_{7D} - h_7$, the optimum ($T_C = 1$ K, $T_{\mathrm{pre3}} = 8$ K: 0.36 MPa, 8.0 kJ/kg), $P_{U,\max} = 1.13$ MPa, $G \approx 7450$ kg m$^{-2}$s$^{-1}$ (from 2.34 mg/s through 20 µm), $Q_C = 18.7$ and 14.7 mW, and about 7 % loss of $Q_C$ per 1 % of $\eta_4$. The He-3 interpolator written then (phase-aware PCHIP, saturation anchors, two-phase by HEM mixing, no extrapolation) was tested against He-4 in CoolProp: on 1500 random off-grid states the median error of $h$ was 0.0016 J/kg (95th percentile 0.16 J/kg), of $\rho$ $6 \times 10^{-7}$ (relative), of $T(P,h)$ $3 \times 10^{-7}$ K; the weakest region was next to the critical point (worst $h$ error 675 J/kg, 0.025 K, at 5.16 K for a critical temperature of 5.195 K), and the cycle $q_C$ agreed with direct CoolProp to about $10^{-4}$. That test checks the interpolation code, not He-3 physics.

## 5.4 Stage 3: the review before this session — problems found

A review of the existing files found issues that directly determined what was changed afterwards:

- **The optimiser did not cap $q_C$ at the latent limit.** It used Eq. 5 directly, which at $\eta = 1$ gives $q_C = \Delta h_T$ even above the optimum pressure, where the cold end cannot be cooled to $T_C$; the symptom is $T_8 < T_C$. In a He-4 stand-in ($T_C = 2.6$ K, $T_{\mathrm{pre}} = 6$ K, HEM): $P_U = 0.3$ MPa gave $T_8 = 5.0$ K and 22.9 mW (valid); $P_U = 0.4$ MPa gave $T_8 = 2.31$ K, below $T_C$, accepted silently at an inflated 70.9 mW; at $P_U \geq 0.5$ MPa the backend failed. *This is the origin of the effectiveness definition of Section 3.3:*

defining $\eta$ against the smaller of the two limits makes $q_C = \min(\Delta h_T, q_L)$ come out of the balance, so the unphysical region cannot be entered.

- $T_8$ **versus the paper's** $T_0 \approx T_C$**.** The paper evaluates the flux at $T_0 \approx T_C$; the code evaluates it at the actual $T_8$ (which is above $T_C$ when $\eta < 1$ or below the optimum). We kept the actual $T_8$ and it is the reason $\dot{m}$ falls with $\eta$ (Section 3.6).
- **Internal pinch not checked.** Only the two ends of each CHEX were tested. This became the 20-point profile check of Section 3.5.
- **Tests.** The interpolator test printed but asserted nothing; it was replaced by unit tests with assertions (29 at present).
- **Smaller points:** HEM grid search was slow (101×101 points); `paper_fit` was valid only for $T_8$ of 1–3 K and 0.2–2 MPa; a plotting dependency was missing from the requirements list.

## 5.5 Stage 4: this session — real data, validation, extensions

When the He-3 tables arrived they were loaded and validated, and the paper was reproduced (27/27). The decisions taken after that, each with its reason:

| Decision | Reason / what prompted it |
|---|---|
| Keep core as black box, one front-door script | the project is driven only from the command line; the browser app that was built was removed at the user's request |
| Effectiveness $= Q/Q_{\max}$ | avoids $T_8 < T_C$ above the optimum (Stage 3); identical to the paper's definition in its regime |
| Optimise $Q_C = \dot{m} q_C$, not only $q_C$ | $q_C$ alone has no interior optimum in $T_{\mathrm{pre,last}}$ (it is a plateau), whereas $Q_C$ does; paper's own Section 5 optimum differs for $q_C$ (0.21 MPa) and $Q_C$ (0.36 MPa) |
| "Practical" optimum within 0.5 % | the $Q_C$ maximum is flat; the warmest $T_{\mathrm{pre}}$ and lowest $P_U$ in the flat region are cheaper |
| Upstream temperatures by Carnot-weighted load, plus a full load map | upstream $T_{\mathrm{pre}}$ cannot change $q_C, \dot{m}, Q_C$ (Section 3.5), so another criterion is needed to pick them |
| Measured flow `mdot_mg_s` accepted | other papers give $\dot{m}$, not an orifice diameter; $d$ and $\dot{m}$ together are rejected as over-specified |
| Effectiveness derived from temperatures where a paper gives them | avoids fitting a free parameter (Liu et al.) |
| Feasibility check before solving | to say *why* a design is impossible ($q_L \le 0$, above inversion, $P_U \le P_D$, ...) instead of returning a number |
| No extrapolation, loud errors | guarantees that a number never comes from outside the tables |
| Named case files, run folders linked by a configuration hash, data-reference block outside the hash | traceability: a result can always be tied to its inputs, and a reference can be added later without orphaning runs |
| A written document instead of slides | the user asked for a document explaining everything in detail |

## 5.6 Cases discussed, and what each taught

1. **Above the optimum pressure (cold-end limit).** Described in Stage 3. Lesson: $q_C = \min(\Delta h_T, q_L)$ is not an extra rule to code, it is what an enthalpy balance with proper limits returns; coding Eq. 5 literally is only valid on the $\Delta h_T$ side.
2. $T_{\mathrm{pre,last}}$**: maximum or plateau?** The handwritten derivation says $q_{C,\mathrm{opt}}$ has a maximum at $T_{\mathrm{pre3}} = 8$ K. Sweeping $T_{\mathrm{pre,last}}$ at the optimal $P_U$ shows $q_C = q_L = 7.98$ kJ/kg for all $T_{\mathrm{pre}} \le 8$ K and a decline above it. So 8 K is the *warmest* point on a plateau, $T_{\mathrm{pre,opt}}$, not a peak. The numbers agree, the interpretation is refined (Section 3.4).
3. **What the upstream precoolers can change.** Sweeping any one upstream $T_{\mathrm{pre}}$ over its full range leaves $q_C$, $\dot{m}$ and $Q_C$ unchanged (total precooling load 26.8 mW constant) and only redistributes the load (Fig. 10).
4. **Orifice model: Γ versus HEM.** Same choking condition, different route; agreement with the paper's Table 2 within 0.25–0.72 % (mean).

5. **Liu et al. (He-4).** With $\eta_3 = 1$ the solver overpredicted by 19 %. The effectiveness was then derived from the paper's own temperatures (17.3 K in, 17.1 K out) giving 0.9857, and the result became 53.1 vs. 52.5 mW. A first value of 0.9971 was a mistake and was corrected: the lesson is that the result is sensitive to $\eta$ and that $\eta$ must come from the paper, not be tuned.
6. **Ma et al. (closed loop).** A large miss (47 vs. 102 mW) was traced, by sensitivity runs, to the evaporator pressure ($P_{\mathrm{sat}}(5.02\,\mathrm{K}) = 0.199$ MPa in the model against 0.16 MPa at the compressor) — i.e. to the no-pressure-drop assumption — not to a numerical fault. The case is kept as an illustration of the model's limit.
7. **Near-critical gap in the tables.** A $Q_C$–$P_U$ sweep failed near 0.1023 MPa because no saturation data exist between 101.7 kPa and the critical pressure; the sweep was restricted to $P_U \geq 0.12$ MPa and failures are recorded as missing values.
8. **Command-line detail.** Values such as `--T_pre -,-,8` were being parsed as options; fixed so that blank entries can be written naturally.

### 5.7 Where the solver differs from the handwritten derivation

1. **Effectiveness:** handwritten (and paper) use $Q_{\max} = h_{A'} - h_C$ only; the solver uses the smaller of the warm-end and cold-end limits. Same results in the paper's regime.
2. **Mass flux:** handwritten uses $G = \Gamma G_{ID}$ with throat from $s^* = s_0$ and $h^* + a^{*2}/2 = h_0$ and a fitted $\Gamma(P_U, T_{\mathrm{pre3}})$; the solver maximises $G(P)$ along the isentrope from the tables (same choking condition, no sound speed, no fit). The paper's Eq. 49 fit remains available.
3. $T_{\mathrm{pre3}} = 8$ **K:** peak (handwritten) versus plateau edge (solver), see case 2.
4. **The CHEX relation** is explicit; the sheets mention solving "implicitly" for the coupled CHEXs, but because the cold boundary is fixed, a single backward sweep from the evaporator suffices.
5. **Additions:** general $n$, other fluids, measured flow, feasibility check, pinch check, upstream map, run storage, validation against other papers.

### 5.8 Ideas considered and deliberately left out

Some ideas from the planning discussions were dropped as too complex, not needed, or a tangent to the objective, and are therefore *not* limitations of this work but scope decisions: an inverse CHEX solver (known $A$ and $B$, find $C$ and $D$); a coupled check of required loads against a real cooler's capacity curves, and the cooler-to-JT inverse case in which loads are prescribed; geometry-based CHEX models; physical PHEX and cryocooler-stage models. The load map (Section 10) is the intended interface to the cryocooler: it states the loads, and the choice of cooler is left to the designer.

## 6 Property models

### 6.1 He-3 from the supplied tables

The paper's results depend on He-3 properties in a region (0.5–3.3 K, 1 Pa–3 MPa) where no standard equation of state package is available, so the solver reads two CSV tables (`data/he3/`): single-phase properties on a $(T, P)$ grid (0.5–300 K, 1 Pa–3 MPa) and saturation properties (0.5–3.2 K). The loader checks columns, units, monotonicity and splits the saturation table into liquid and vapour branches; `python jt_run.py check` reports the SHA-256 of both files so a run can be tied to a data version (here `b17901e3e299` and `8a674a6a03da`) and confirms $P_{\mathrm{sat}}(1\,\mathrm{K}) = 1160.3$ Pa, which is the paper's 1.16 kPa.

### 6.2 Interpolation

- **Method.** $h(P,T)$, $s(P,T)$, $\rho(P,T)$ are interpolated with monotone piecewise-cubic Hermite polynomials (PCHIP), first along each tabulated isotherm in $\ln P$, then across isotherms in $T$ at the requested pressure. PCHIP does not overshoot: crucial for $c_p = \partial h/\partial T$ where $c_p$ has a sharp peak (Fig. 3).
- **Phase classes.** The table spans the liquid and vapour phases. For each $(P,T)$ the phase is decided by comparing $T$ with $T_{\mathrm{sat}}(P)$; interpolation is done only among nodes of the same class so that the discontinuity at saturation is not smoothed over. Above the critical point (3.3246 K, 116.4 kPa) a single class is used; the critical point is used for *classification*, closing the saturation dome.

- **Near-critical gap.** Between 101.7 kPa (last tabulated saturation point) and the critical pressure the tables have no saturation data; requests there raise `OutOfRange`. This is why the pressure sweeps of the $Q_C$ study start at 0.12 MPa. A liquid-class lookup needs at least three nodes, hence it is not available below $\approx 45$ Pa.
- **No extrapolation.** Anything outside the tables raises `OutOfRange`; $T_C$ is limited to $\leq 3.2$ K. This is deliberate (Section 4, item 5).
- **Inverse.** $T(P, h)$ is obtained by bracketed root finding on $h(P, T)$; a $T \to h \to T$ round trip at 0.36 MPa, 8 K returns 8.00000 K (part of the data check). Absolute values of $h$ and $s$ depend on the table's reference state; only differences enter any result, so the reference state is not checked.

### 6.3 Other fluids

For He-4, Ne, $H_2$, para-$H_2$, $N_2$, Ar, $O_2$, $CH_4$ and others, the same interface (`h, s, rho, Psat, T_from_Ph, ...`) is served by CoolProp. Two restrictions follow from the tables-free approach: the high pressure must exceed the critical pressure of the fluid (He-4 0.227 MPa is not an issue, but Ne 2.66, $H_2$ 1.30, $N_2$ 3.40 MPa are), otherwise the hot stream condenses inside the exchangers and the property lookup fails; and the feasibility check restricts $T_C$ from the triple point to $0.995\,T_{\mathrm{crit}}$.

## 7 Implementation

### 7.1 Structure

| File | Content |
|---|---|
| `jt/properties.py` | `Helium3Fluid`: CSV-backed tables, PCHIP interpolation, saturation, inverse, error classes |
| `jt/cycle_core.py` | state points, the CHEX solver (Eqs. 1–5), `solve_cycle`, the HEM choked flux (Eq. 14), the CoolProp adapter, flags |
| `jt/design_optimize.py` | last-stage optimisation of $P_U$, $T_{\mathrm{pre,last}}$; Eq. 49 fit; case-file naming |
| `jt/phex_load_map.py` | PHEX loads versus an upstream temperature |
| `jt/workflow.py` | configuration normalisation, feasibility check, optimiser pipeline, upstream search, run storage |
| `jt_run.py` | the command-line front door |
| `run_all.py` | one-command validation chain (data check, paper, other papers) |
| `reproduce_zhang_dang_2021.py` | reproduction of the paper's figures/fits |
| `tests/test_workflow.py` | 29 unit tests |

### 7.2 Solving a cycle (`solve_cycle`)

Given $T_C, P_U, T_H$, $T_{\mathrm{pre},1..N-1}$, $\eta_{1..N}$ and $d$ or $\dot{m}$:

1. Compute $P_D = P_{\mathrm{sat}}(T_C)$ and the evaporator outlet $h_{10} = h_g(T_C)$.
2. **Last CHEX first.** Its hot inlet $(T_{\mathrm{pre,last}}, P_U)$ and cold inlet $(h_{10})$ are known; apply Eqs. (1)–(5) to get $Q$, hence $h_8$, $h_{11}$ and the states on both streams. $q_C = h_{10} - h_8$.
3. **Work upstream.** The cold-stream inlet of CHEX $i$ is the outlet of CHEX $i+1$ (known). The hot inlet of CHEX $i$ is $(T_{\mathrm{pre},i-1}, P_U)$ (or $T_H$ for $i = 1$). Apply Eqs. (1)–(5). The PHEX load follows from the enthalpies on either side: $q_{\mathrm{pre},i} = h(\text{CHEX } i \text{ hot out}) - h(P_U, T_{\mathrm{pre},i})$.
4. $\dot{m} = \pi d^2 G/4$ from Eq. (14) at the stagnation state 8 (or the given $\dot{m}$); $Q_C = \dot{m} q_C$, $Q_{\mathrm{pre},j} = \dot{m} q_{\mathrm{pre},j}$.
5. **Checks.** global balance Eq. (13), non-negative PHEX loads, temperature ordering, internal pinch scan, choking.

Because every step is explicit (no iteration on the cycle), a solve takes milliseconds, which makes the scans of Section 10 cheap; the cost is in the property lookups and the choked-flux search.

## 7.3 Theory-to-code map

| Equation / statement | Paper | Where in the solver |
|---|---|---|
| CHEX energy balance, limits (1)–(3) | Eqs. 2, 5, 9–12 | CHEX solver in `cycle_core.py` |
| Effectiveness (5) (differs from Eq. 4 beyond optimum) | Eq. 4 | CHEX solver, `eta` |
| $q_C = \min(\Delta h_T, q_L)$ (9) | Eqs. 30–31 | `solve_cycle` (emerges, not coded); tested by the scorecard check "$q_C = \min(\Delta h_T, q_L)$" |
| Optimum $\Delta h_T = q_L$ (11) | Eq. 34 | `design_optimize.py` (scan, then root refinement) |
| Fits $P_{U,\mathrm{opt}}$, $q_{C,\mathrm{opt}}$ | Eqs. 35–36, Table 1 | compared in `reproduce_zhang_dang_2021.py` |
| Cascade balance (13) | Eq. 22 | global-balance check in `solve_cycle` |
| HEM flux (14) | Eqs. 41–48 | `cycle_core.py` |
| Paper's flux fit | Eq. 49, Table 2 | `design_optimize.py`, `--flow-model paper_fit` |
| $Q_C = \dot{m} q_C$ | Eq. 50 | `solve_cycle` |
| Upstream criterion (15) | — (new) | `workflow.optimise_upstream` |
| Upstream sensitivity map | — (new) | `workflow.upstream_map` |
| Feasibility | — (new) | `workflow.check_config` |

## 7.4 Feasibility check

`jt_run.py feasible case.json` answers "is this design physically possible and why not". In order: (1) $T_C$ inside the fluid window (He-3 0.5–3.2 K; CoolProp fluids triple point to $0.995 T_{\mathrm{crit}}$); (2) $P_U > P_D$; (3) $T_{\mathrm{pre}}$ strictly decreasing from $T_H$ and all above $T_C$; (4) $q_L = h_g(T_C) - h(P_U, T_C) > 0$, otherwise no refrigeration at any $T_{\mathrm{pre}}$ (the message gives $P_{U,\max}$); (5) $\Delta h_T > 0$ at $T_{\mathrm{pre,last}}$, otherwise the gas is above its inversion temperature; (6) a warning if $T_{\mathrm{pre,last}} > T_{\mathrm{pre,opt}}$ (then $q_C = \Delta h_T$ is limited by the precooling); (7) a trial solve whose flags are classified as errors (a PHEX would have to heat the gas, balance fails, no net refrigeration) or warnings (internal pinch, unchoked flow). The verdict is *feasible*, *feasible with warnings* or *infeasible*.

## 7.5 Run storage, naming and the data reference

- **Case names.** `<Fluid>_<T_C>K_<d>um_<N>_stage[_eta<x>]`, e.g. `Helium_3_1K_20um_4_stage`; a measured flow replaces the orifice by `<mdot>mgs`, `nod` marks "no flow information", and `_v2`, `_v3` are appended if the name already exists with different content. He-4 is labelled `Helium_4`.
- **Run folders.** Every `run` writes `results/runs/<config name>/<stamp>_<solve|optimize>/` with `config.json`, `meta.json`, `result.json`, `states.csv`, `ph_cycle.png` and, for optimisation, `map_qC.png`, `map_QC.png`, `stage_curves.png`, `upstream_map.png/csv`.
- **Linking runs to configurations.** `meta.json` holds a hash of the input configuration (fluid, $T_C$, $P_U$, $T_H$, $T_{\mathrm{pre}}$, $\eta$, $d$ or $\dot{m}$, optimiser settings). The data-reference block (citation, DOI, location in the paper, kind, note) is *excluded* from the hash, so a reference can be added or corrected later without orphaning the runs. `python jt_run.py runs` lists the runs against configurations.

## 7.6 Command reference

| Command | Purpose |
|---|---|
| `check` | He-3 tables present and consistent |
| `validate [-quick]` | workflow 1: data check, tests, paper reproduction, scorecard |
| `new -fluid ...` | create a named case with data reference, check feasibility |
| `feasible cases/x.json` | feasibility verdict and reasons |
| `run cases/x.json [-optimize ...]` | solve, or optimise if anything is blank |
| `runs` | list saved runs and their configurations |
| `solve cases/*.json` | plain solve with reported-value comparison and sweeps |
| `optimize, phex` | original last-stage optimiser and PHEX load map |
| `python run_all.py [-quick]` | everything in one command |

Blank inputs (- for $T_{\mathrm{pre}}$ or $P_U$) trigger the optimiser; the options are `--drive Q_C|Q_C_practical|q_C`, `--flow-model hem|paper_fit`, `--pu-min/--pu-max`, `--tpre-min/--tpre-max`, `--hem-grid`, `--save-final`.

### 7.7 Where the generated files go

| Location (relative to the project folder) | Content |
|---|---|
| `results/validation/` | paper reproduction report, scorecard (`scorecard.json`), figures |
| `results/validation_all/` | `run_all.py` summary (`summary.csv`, `summary.json`), per-case results |
| `results/cases/<name>/` | output of `solve` (result, states, sweeps) |
| `results/runs/<config>/<stamp>_<kind>/` | output of `run` (Section above) |
| `cases/` | case files (inputs, references, reported values) |
| `paper_figs/ (delivered with this document)` | the figures and data used here |

`results/` is generated output and can be deleted at any time.

## 8 Validation I: Zhang & Dang (2021)

All numbers are produced by `python jt_run.py validate` (27 checks, all pass) and by additional runs listed below. Inputs are the paper's: $T_C = 1$ K, $T_H = 300$ K, $d = 20$ µm, $T_{\mathrm{pre}} = (80, 25, 8)$ K (for the load table), He-3 tables.

### 8.1 Last-stage optimum (Figs. 5–8 of the paper)

| Quantity | Solver | Paper | Difference |
|---|---|---|---|
| $P_{U,\mathrm{opt}}$ at $T_{\mathrm{pre3}} = 8$ K [MPa] | 0.3584 | 0.36 | −0.4 % |
| $q_{C,\mathrm{opt}}$ [kJ/kg] | 7.979 | 8.0 | −0.3 % |
| $P_{U,\max}$ ($q_L = 0$) [MPa] | 1.131 | 1.13 | +0.1 % |
| $P_U$ for max $q_C$, $T_{\mathrm{pre}} = 5$ K [MPa] | 0.213 | 0.21 | +1.4 % |
| $P_U$ for max $Q_C$, $T_{\mathrm{pre}} = 5$ K [MPa] | 0.352 | 0.36 | −2.2 % |
| Eq. (11) satisfied at the optimum (max rel. gap) | $4 \times 10^{-8}$ | — | consistency |
| $q_C = \min(\Delta h_T, q_L)$ (max abs. deviation) | $7 \times 10^{-15}$ | — | consistency |

The last two lines test that our implementation *reproduces the paper's analytical result from first principles*: the minimum rule and the optimum condition emerge from the energy balance, they are not coded in.

**Fits of the optimum (paper Eqs. 35–36, Table 1).** Over the paper's range of $T_{\mathrm{pre3}}$ our $P_{U,\mathrm{opt}}$ differs from the paper's fit by 1.5 % on average and 3.7 % at worst; $q_{C,\mathrm{opt}}$ by 3.3 % on average; in the $T_{\mathrm{pre3,opt}}$ versus $P_U$ relation (paper Fig. 7) the differences at 0.3, 0.5, 0.7, 0.9 MPa are $0.01, 0.23, 0.34, 0.78$ K. No optimum fell on the edge of the search range.

### 8.2 Mass flux (paper Eqs. 41–49, Table 2)

| $T_C$ [K] | 1 | 2 | 2.6 | 3 |
|---|---|---|---|---|
| HEM flux vs. paper Table 2, mean error [%] | 0.25 | 0.43 | 0.56 | 0.72 |
| Eq. 49 form vs. our HEM data, worst error [%] | 0.18 | 0.43 | 0.76 | 1.35 |

**Table 1:** Agreement of the HEM mass flux with the paper's real-gas flux.

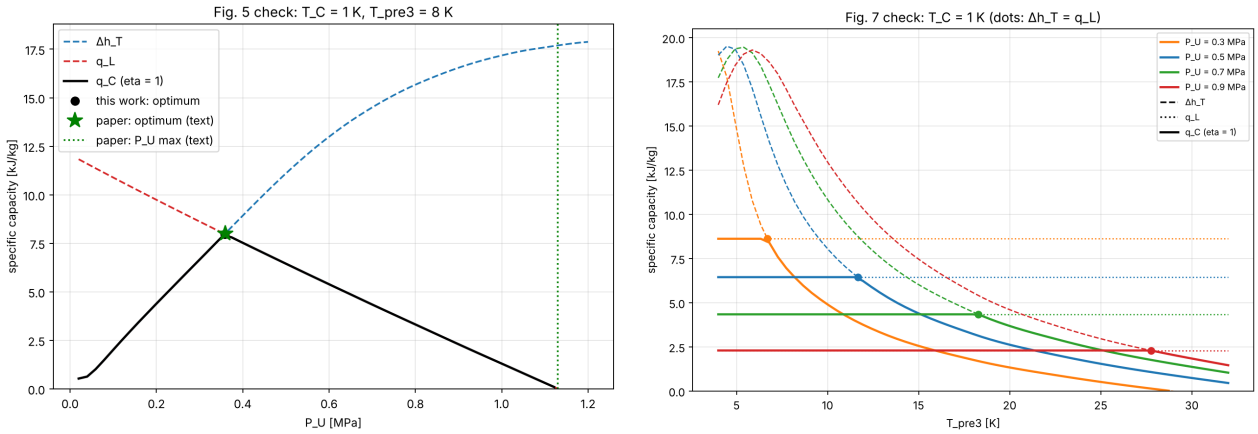



**Figure 6:** Reproduction of the paper's Fig. 5 ($q_C$ vs. $P_U$ at $T_{\mathrm{pre3}} = 8$ K) and Fig. 7 ($q_C$ vs. $T_{\mathrm{pre3}}$ for several $P_U$).

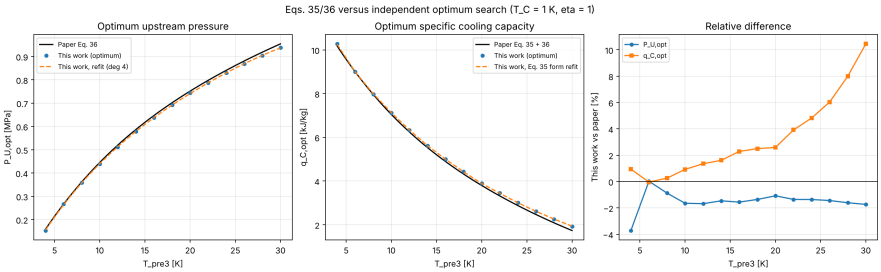



**Figure 7:** Solver optima against the paper's fits, Eqs. (35)–(36).

## 8.3 Effect of the CHEX efficiency (paper Figs. 14–16)

Keeping all other CHEXs ideal and varying the last effectiveness (optimal $P_U = 0.358$ MPa, $T_{\mathrm{pre3}} = 8$ K, $d = 20$ µm):

| $\eta_4$ | $T_8$ [K] | $T_{11}$ shortfall [K] | $q_C$ [kJ/kg] | $\dot{m}$ [mg/s] | $Q_C$ [mW] |
|---|---|---|---|---|---|
| 1 | 1.000 | 0 | 7.961 | 2.339 | 18.62 |
| 0.995 | 1.180 | 0.043 | 7.718 | 2.332 | 18.00 |
| 0.99 | 1.349 | 0.078 | 7.475 | 2.322 | 17.36 |
| 0.98 | 1.651 | 0.149 | 6.989 | 2.297 | 16.06 |
| 0.97 | 1.910 | 0.219 | 6.503 | 2.267 | 14.74 |
| 0.95 | 2.333 | 0.360 | 5.531 | 2.197 | 12.15 |
| 0.90 | 3.086 | 0.712 | 3.102 | 1.994 | 6.19 |

("$T_{11}$ shortfall" is $T_7 - T_{11}$, the temperature by which the return stream leaves the CHEX colder than the warm-end temperature.) Comparison with the paper: $Q_C(\eta = 1) = 18.62$ vs. 18.7 mW, $\dot{m} = 2.339$ vs. 2.34 mg/s, $q_C = 7.96$ vs. 8.0 kJ/kg; $Q_C(\eta = 0.97) = 14.74$ vs. 14.7 mW and a relative drop of 20.8 % vs. about 21 % (paper Fig. 16). The paper's statement that for $\eta_4 \geq 0.97$ the temperature differences at the cold and warm ends should not exceed 1 K and 0.2 K (its Fig. 14) is met: we find 0.91 K ($T_8 - T_C$) and 0.22 K ($T_7 - T_{11}$) at $\eta = 0.97$. The sensitivities of $Q_C$ are about $-7$ % per 0.01 of $\eta$ near $\eta = 1$ and the change is almost linear (paper Fig. 16); the contribution of the flux ($\dot{m}$ drops by 3 %) is small compared with that of $q_C$ (18 %) consistent with Eq. (12).

## 8.4 Precooling loads for the paper's temperatures

At $P_U = 0.36$ MPa and $T_{\mathrm{pre}} = (80, 25, 8)$ K the solver gives $Q_{\mathrm{pre}} = (0.511, 9.698, 16.613)$ mW and $Q_C = 18.62$ mW, $\dot{m} = 2.339$ mg/s; i.e. the 8 K stage has to absorb about 16.6 mW. Cascade balance Eq. (13) is satisfied to rounding. The solver flags an internal temperature crossover in CHEX 1 (min. gap $-0.013$ K) and the unchoked flow (the inlet is liquid-like at 8 K, so the HEM maximum is at the back pressure; the flux is still the HEM maximum).

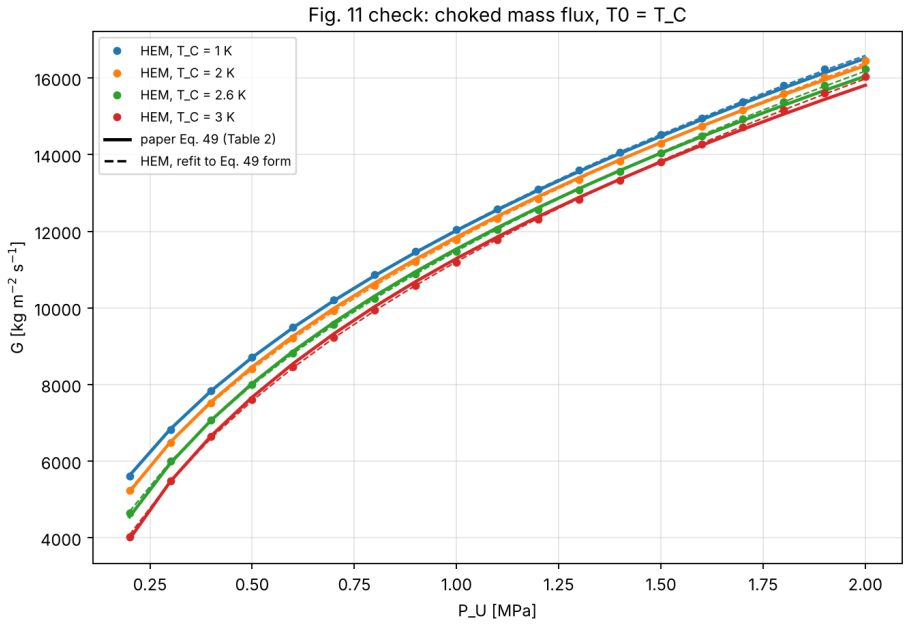



**Figure 8:** HEM mass flux versus $P_U$ for several $T_C$ with the paper's fitted curves (cf. its Fig. 11).

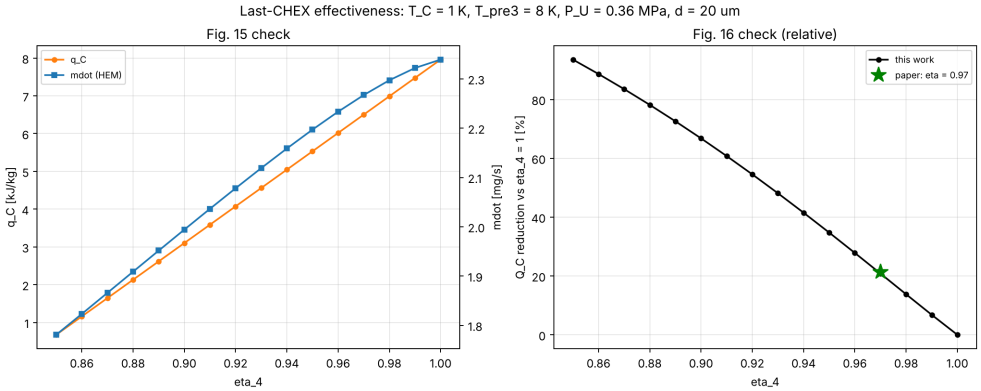



**Figure 9:** Reproduction of the paper's Figs. 15 and 16: effect of $\eta_4$ on $q_C$, $\dot{m}$ and $Q_C$.

### 8.5 Summary of paper validation

27 of 27 scorecard checks pass. The scorecard separates *anchors* (numbers read from the paper), *consistency* checks (our implementation against its own identities) and *model* checks (HEM versus real-gas flux, a miss is only a warning). The tolerances are provisional and wider than the agreement obtained (the largest anchor deviation is 2.2 %); tightening them to 2–3 % is possible if wanted.

## 9 Validation II: other papers with the same architecture

The solver, unchanged, was applied to five cases from two further papers with the same coupled architecture, using the paper's inputs only. They differ from the He-3 paper in two respects that matter: they are *measurements* or design values of a real cooler (not an ideal model), and the working fluid is He-4 (CoolProp). Each case is a file in `cases/` with a `reference` block (citation, DOI, location, assumptions) and the paper's values under `reported`.

### 9.1 Liu et al. (2017): He-4 open-cycle JT cooler

*Int. J. Heat Mass Transf.* **109** (2017) 1094. Inputs (their Sect. 2.2/Eq. 1): $T_a = 17.3$ K, 1.137/0.104 MPa, $\dot{m} = 9.1$ mg/s, calculated $Q_{\max} = 52.5$ mW, measured 48.9–49.4 mW. $T_C = T_{\mathrm{sat}}(0.104\,\mathrm{MPa}) = 4.2517$ K. The paper does not give an effectiveness; it gives the temperature of the supply entering the last exchanger

| Case | $T_C$ [K] | $P_U$ [MPa] | $\dot{m}$ [mg/s] | $T_{\mathrm{pre}}$ [K], $\eta$ |
|---|---|---|---|---|
| Liu 2017, open cycle | 4.2517 | 1.137 | 9.1 | (60$^\dagger$, 17.3); (1$^\dagger$, 1$^\dagger$, 0.9857) |
| Ma 2019 design | 4.5 | 2.0 | 10.5 | (100, 15); 0.97 each |
| Ma 2019 open, 4×20 µm | 4.55 | 2.0 | 11.08 | (100, 15); 0.97 each |
| Ma 2019 closed, 4×20 µm | 5.02 | 2.0 | 11.02 | (100, 15); 0.97 each |
| Ma 2019 Sci China | 2.71 | 0.6 | 1.2 | (64.5, 14.6); (0.972, 0.973, 0.972) |

**Table 2:** Inputs of the other-paper cases ($^\dagger$ = placeholder not given by the paper; does not affect $q_C$, $\dot{m}$ or $Q_C$, only the load split).

| Case | $Q_C$ solver [mW] | $Q_C$ paper [mW] | Difference |
|---|---|---|---|
| Liu 2017 (calculated $Q_{\max}$) | 53.1 | 52.5 | +1.1 % |
| Ma 2019 design | 90.2 | 100 | −9.8 % |
| Ma 2019 open (measured) | 92.4 | 100.3 | −7.9 % |
| Ma 2019 closed (measured) | 47.2 | 102.3 | −53.9 % |
| Ma 2019 Sci China (measured) | 4.50 | 1.48 | +204 % |

**Table 3:** Gross refrigeration at the cold end, solver versus paper.

(17.3 K) and of the return leaving it (17.1 K), so $\eta_3$ follows from Eq. (16): 0.9857. The upstream values (60 K, $\eta_1 = \eta_2 = 1$) are placeholders. **Result:** $q_C = 5.835$ kJ/kg, $Q_C = 53.1$ mW (paper 52.5, +1.1 %) and $P_D = 104$ kPa. With $\eta_3 = 1$ the solver would give about 19 % more, so *the result is sensitive to the derived effectiveness* and the agreement is only meaningful because $\eta_3$ is derived from the paper's own temperatures, not fitted. The 3.6 mW between the paper's calculation and measurement is the heat in-leak (A2) that the model excludes.

## 9.2 Ma et al. (2019), Cryogenics 104: 100 mW @ 5 K closed-loop JT cooler

Three cases (design, open-loop and closed-loop tests), all with precool (100, 15) K, 2.0 MPa and four-orifice element; $\eta$ is bounded by the paper ("not lower than 0.97") rather than measured, so $\eta = 0.97$ is an assumption; the measured flows are used because the multi-orifice element is not modelled.

- **Design.** $Q_C = 90.2$ mW (paper 100 mW), $Q_{\mathrm{pre}} = (207, 278)$ mW (design 350 and 260 mW). The sensitivity to $\eta$ is large: 90.2 mW at $\eta = 0.97$ and 108.8 mW at $\eta = 1$; the paper's 100 mW lies at $\eta \approx 0.985$, inside the bounds given. The second precooler load agrees (+7 %); the first does not (−41 %): upstream loads depend on the placeholder precooling temperature and on parasitic loads that an ideal model has no access to.
- **Open loop (measured).** $Q_C = 92.4$ mW vs. 100.3 mW (−7.9 %); the precool temperatures of the test are not given, so the design ones are used.
- **Closed loop (measured).** $Q_C = 47.2$ mW vs. 102.3 mW. The paper quotes $P_D = 0.16$ MPa at the compressor suction; our evaporator pressure is $P_{\mathrm{sat}}(5.02\,\mathrm{K}) = 0.199$ MPa because the model has no pressure drop between evaporator and compressor, and at that temperature ($T_C$ close to the He-4 critical temperature 5.19 K) $q_L$ is already small. Sensitivities: 78 mW with $T_C = T_{\mathrm{sat}}(0.16\,\mathrm{MPa}) = 4.745$ K; 66 mW with $\eta = 1$ at $T_C = 5.02$ K. The case thus shows that *the model cannot be applied to a measured closed loop without a return-line pressure drop*, which is a limit of A3, not a numerical problem.

## 9.3 Ma et al. (2019), Sci. China Tech. Sci. 62: 2.65 K closed-loop hybrid JT cooler

Inputs from their Table 2 and text: precool (64.5, 14.6) K, $P_h = 0.6$ MPa, $\eta = (0.972, 0.973, 0.972)$, $\dot{m} = 1.2$ mg/s, measured 1.48 mW at 2.71 K, return pressure 12 kPa at the compressor (13.3 kPa at the evaporator). Result: $Q_C = 4.50$ mW and $P_D = 15.1$ kPa (the saturation pressure of He-4 at 2.71 K). The solver is optimistic by a factor 3, as the model has no heat leak: at 1.2 mg/s and $q_C = 3.75$ kJ/kg the ideal refrigeration is 4.5 mW while the cooler's net refrigeration is 1.48 mW, i.e. about 3 mW of parasitic load (not given by the authors, inferred only as the difference). Low-temperature, low-power coolers are dominated by parasitics, and the model, correctly, gives an upper bound.

### 9.4 Summary of the cross-paper comparison

- Where the source paper reports an *ideal-model* number from the same assumptions (Zhang & Dang; Liu's calculated $Q_{\max}$) the solver agrees to 0.4–2 %.
- Where the source is a *measurement* the solver gives the model upper bound and the discrepancy is explained by what the model leaves out: heat leak (Sci China case), a return-line pressure drop (closed-loop case), the assumed $\eta$ (design case: ±9 % inside the paper's own bounds).
- The case files carry sensitivities (in `reference.note`) so the reader can see which assumption drives each discrepancy.
- NIST and Shimazaki cooler cases were not built because the available papers do not give enough inputs ($T_{\text{pre}}$, $\eta$ and $\dot{m}$ or $d$); Narasaki's references (25, 29) were not supplied.

In `run_all.py` the idealised comparisons (Liu $Q_C$ 3 %, $P_D$ 1 %; Ma design $Q_C$ 12 %, $P_D$ 1 %, $Q_{\text{pre},2}$ 10 %) are scored PASS/FAIL, the measured ones are informative (INFO) only.

## 10 Application: design of a coupled hybrid JT cooler

The validated solver is used the way a designer would use it: choose the target, find the operating point, and read off what the precooling cryocooler has to supply.

### 10.1 Procedure

1. Create a named case: `python jt_run.py new --fluid he3 --T_C 1.0 --eta 1,1,1,1 --T_pre -,-,-,8 --P_U - --d_um 20`. This produces `Helium_3_1K_20um_4_stage.json`. (Here the last stage temperature is given, 8 K, and the rest is left to the optimiser.)
2. `python jt_run.py feasible cases/Helium_3_1K_20um_4_stage.json`: verdict, $P_{U,\max} = 1.131$ MPa, $T_{\text{pre,opt}}$.
3. `python jt_run.py run cases/Helium_3_1K_20um_4_stage.json`: optimise and solve.

### 10.2 Result for the paper's cooler

The optimiser finds $P_U = 0.3584$ MPa and $Q_C = 18.62$ mW (both for $q_C$ and $Q_C$), with $T_{\text{pre,opt}} = 8$ K at that pressure, without being told the optimum; it then selects the upstream temperatures by criterion (15):

| PHEX | Gas in [K] | Gas out ($T_{\text{pre}}$) [K] | $q$ [kJ/kg] | $Q$ [mW] |
|---|---|---|---|---|
| 1 | 26.24 | 25.64 | 4.240 | 9.90 |
| 2 | 12.77 | 12.29 | 3.564 | 8.32 |
| 3 | 8.446 | 8.00 | 3.665 | 8.56 |
| Total precooling | | | 11.47 | 26.77 |

The refrigeration $q_C = 7.979$ kJ/kg, $\dot{m} = 2.334$ mg/s (the small difference from 2.339 is the optimal $P_U$ 0.3584 vs. 0.36 MPa). The state table is printed by `run` and stored in `states.csv`; $T_8 = 1$ K, $T_{11} = 8$ K, $h_8 = 5.13$ kJ/kg.

### 10.3 The upstream map: how to select the precooling cryocooler

Because the upstream temperatures only redistribute a fixed total of 11.47 kJ/kg (26.8 mW), the useful design information is the *map* of loads against temperature. Figure 10 sweeps PHEX 1 (left) and PHEX 2 (right) with the others fixed. Reading it:

- Total precooling load is constant (dashed black).
- Raising $T_{\text{pre},1}$ moves load from PHEX 1 to PHEX 2; above $\approx 90$ K PHEX 1 carries almost nothing (the exchanger 1 already brings the gas to that temperature) and PHEX 2, the second stage, carries about 18 mW.
- PHEX 2 at 8–25 K trades load with PHEX 3: a colder $T_{\text{pre},2}$ moves load from PHEX 3 to PHEX 2.

The Carnot-weighted cost $\Phi$ (Eq. 15) is 263 at the chosen $T_{\text{pre},1} = 26.7$ K (this is the minimum of the sweep; 316 at 12 K, 305–316 above 70 K) and 264 at $T_{\text{pre},2} = 11.1$ K. For a real cryocooler the capacity curve decides:

choose the cooler whose stage capacities at its stage temperatures meet or exceed the loads in this map. For the paper's cooler a four-stage cryocooler with stages near 26, 12 and 8 K carrying about 10, 8 and 9 mW (plus the first stage at 80–300 K carrying no significant load) satisfies the optimum.

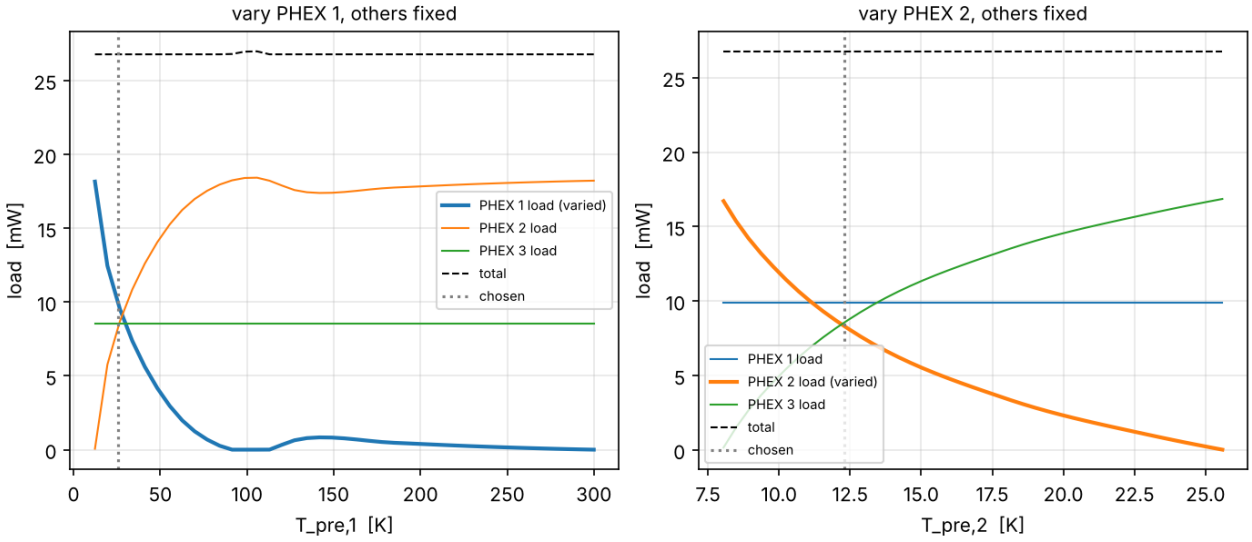



**Figure 10:** Loads of the three PHEXs when PHEX 1 (left) or PHEX 2 (right) temperature is varied; total constant; the dotted line marks the chosen value.

### 10.4 Using the same machinery for another fluid or target

The same three commands work for any supported fluid: e.g. He-4 at 4.5 K, or Ne. The feasibility check will report $P_{U,\max}$, the inversion limit and $T_{\mathrm{pre,opt}}$, so one can see directly whether a target is possible and how warm the last-stage precooler can be, before any optimisation. The five cases of Section 9 can be re-run with `--optimize` to see what $T_{\mathrm{pre}}$ and $P_U$ would be best for those coolers.

## 11 Limitations

- **No heat leak, no pressure drop, no axial conduction,** $C_d = 1$ (A2, A3, A9): the solver is an *upper bound* on a real cooler. The Sci China and closed-loop cases show the size of the effect.
- **Effectiveness is an input.** The model does not predict $\eta$ from geometry; the result is very sensitive to it (about $-7\,\%$ in $Q_C$ per 0.01 of $\eta_{\mathrm{last}}$), so $\eta$ must come from measurement or from a separate exchanger model.
- **Precooling is imposed.** Cryocooler capacity curves are not modelled; the load map tells which capacities are needed.
- **Mass flow** is the choked ideal-orifice flow or a given value; no compressor coupling, no multi-orifice element.
- **He-3 range:** $T_C \leq 3.2$ K, no extrapolation, near-critical gap (101.7–116.4 kPa); the liquid class is unavailable below about 45 Pa.
- **Other fluids:** $P_U$ must exceed the critical pressure; CoolProp accuracy is that of its equations of state.
- **Internal pinch** is flagged but not corrected (the first CHEX of the paper's design shows a $-0.01$ K crossover).
- **Upstream criterion** (Carnot weighting) is one choice; the map should be used with the actual cryocooler.
- **Scorecard tolerances** are provisional (looser than the agreement obtained).
- **Not validated:** NIST and Shimazaki coolers (inputs not available); mixtures; two-phase inlet conditions.

## 12 Reproducibility

1. `pip install numpy scipy pandas matplotlib CoolProp`; place the He-3 tables in `data/he3/`.
2. `python run_all.py --quick` (about 80 s): data check, paper reproduction and scorecard (27/27), other-paper cases (summary in `results/validation_all/`). `--skip-paper1` runs only the other papers.

3. `python jt_run.py validate` (full grids) and `python -m unittest discover -s tests -t .` (29 tests).
4. The figures of this document are in `paper_figs/`: `fig5`, `fig7`, `fits_eq35_36`, `flux`, `fig15_16` (paper reproduction), `cp_He3`, `QC_vs_PU_Tpre` (analysis), `paper_case_ph_cycle`, `liu_ph_cycle`, `ma_design_ph_cycle`, `ma_scichina_ph_cycle` (cycles on the $P$–$h$ plane), `design_stage_curves`, `design_upstream_map` (design example), with the data in `scorecard.json`, `reproduction_report.json`, `extras.json`, `design_upstream_map.csv`.

# A State labelling for general $N$

For $N$ CHEXs the hot-stream states are numbered $1, \ldots, 2N$ in order along the supply path (CHEX $i$ hot inlet $2i-1$, hot outlet $2i$; the following PHEX leaves state $2i+1$), orifice outlet $2N+1$, evaporator outlet $2N+2$, then the return-stream states $2N+3, \ldots, 4N+2$ from the cold end of CHEX $N$ to the warm exit. For $N = 4$: 1–8 hot, 9 orifice outlet, 10 evaporator outlet, 11–14 return (the paper's numbering, its Fig. 2). The `states.csv` file holds $T, P, h$, phase and a role string for each.

# B Flags

| `Flag` | Meaning |
|---|---|
| `INTERNAL_TEMPERATURE_CROSSOVER` | stream temperatures cross inside a CHEX (warning) |
| `UNCHOKED_RESTRICTION_FLOW` | HEM maximum flux at the back pressure; flux still the HEM maximum (warning) |
| `(error class)` | a PHEX would have to heat the gas; balance fails; no net refrigeration |

# C Example case file

```
{ "name": "Liu 2017 He-4 open-cycle JT, last stage at 17.3 K",
  "fluid": "Helium", "T_C": 4.2517, "P_U": 1.137, "T_H": 300,
  "T_pre": [60.0, 17.3], "eta": [1.0, 1.0, 0.9857],
  "mdot_mg_s": 9.1,
  "reference": {"citation": "...", "location": "Sec. 2.2 / Eq. (1)",
                "kind": "...", "note": "..."},
  "reported": {"Q_C_mW": 52.5, "P_D_kPa": 104.0} }
```

Keys beginning with `_` are ignored; `d_um` and `mdot_mg_s` are mutually exclusive.

# D Scorecard of the paper reproduction

| Check | Solver | Paper | Tolerance |
|---|---|---|---|
| $P_{U,\mathrm{opt}}$ at $T_{\mathrm{pre3}} = 8$ K [MPa] | 0.3584 | 0.36 | $\pm 0.03$ |
| $q_{C,\mathrm{opt}}$ [kJ/kg] | 7.979 | 8.0 | $\pm 0.4$ |
| $P_{U,\max}$ [MPa] | 1.131 | 1.13 | $\pm 0.05$ |
| $q_C = \min(\Delta h_T, q_L)$ | $7 \times 10^{-15}$ | — | $\leq 10^{-6}$ |
| $\Delta h_T = q_L$ at optimum (Eq. 34) | $3.7 \times 10^{-8}$ | — | $\leq 0.01$ |
| $P_{U,\mathrm{opt}}$ vs. Eq. 36, mean / worst [%] | 1.50 / 3.72 | — | 5 / 10 |
| $q_{C,\mathrm{opt}}$ vs. Eq. 35+36, mean [%] | 3.26 | — | 5 |
| $T_{\mathrm{pre3,opt}}$ at 0.3/0.5/0.7/0.9 MPa [K] | 0.01/0.23/0.34/0.78 | Eq. 36 | $\pm 1.5$ |
| HEM flux at $T_C = 1, 2, 2.6, 3$ K, mean [%] | 0.25/0.43/0.56/0.72 | Table 2 | 15 |
| Eq. 49 form, worst [%] | 0.18/0.43/0.76/1.35 | — | 2 |
| $q_C$ ($\eta = 1$) [kJ/kg] | 7.961 | 8.0 | $\pm 0.4$ |
| $Q_C$ ($\eta = 1$) [mW] | 18.62 | 18.7 | $\pm 1$ |
| $\dot{m}$ ($\eta = 1$) [mg/s] | 2.339 | 2.34 | $\pm 0.15$ |
| $Q_C$ ($\eta = 0.97$) [mW] | 14.74 | 14.7 | $\pm 1$ |
| $Q_C$ drop at $\eta = 0.97$ [%] | 20.82 | 21.39 | $\pm 3$ |

27 checks in total (the table groups related ones); all pass.

## Reference

T. Zhang and H. Dang, "Investigations on a 1 K hybrid cryocooler composed of a four-stage Stirling-type pulse tube cryocooler and a Joule–Thomson cooler. Part A: Theoretical analyses and modeling", *Cryogenics* **116** (2021) 103282. Liu et al., *Int. J. Heat Mass Transf.* **109** (2017) 1094–1099. Ma et al., *Cryogenics* **104** (2019) 102983. Ma et al., *Sci. China Tech. Sci.* **62** (2019) 361–364.